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国家自然科学基金 资助项目结题/成果报告

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项目名称：	蜂窝车联网中边缘智能的协同卸载与隐私保护技术研究		
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项目摘要

中文摘要:

蜂窝车联网 (C-V2X) 搭乘5G及下一代通信技术, 逐渐呈现应用与移动边缘计算 (MEC) 新技术紧密结合的发展态势。然而, 由于车辆的高速移动, MEC使能技术推动C-V2X网络应用落地还存在一系列关键技术阻碍。本课题将MEC平台部署在靠近车辆用户的路侧, 融合联邦学习框架, 解决车辆智能网联中不同层面不同类别业务的实际应用。从模型压缩、条件计算和异步算法设计三方面尝试将深度学习框架引入到“车-路-云”服务架构, 形成纵向融合和横向协作的自适应协同; 针对车辆服务请求的个性化和服务器的资源异构性, 分别提出“车-车”、“车-路”、“车-路-云”场景下的任务卸载方法, 利用智能算法的优势增强计算环境的服务能力; 提出基于博弈的激励与竞价机制, 细粒度优化多服务器供给的竞争环境; 最后, 赋能联邦学习, 建立区块链与联邦学习的融合方案, 提出一种实现信息共享的分层区块链联合学习算法, 实现C-V2X网络中多属性数据的安全共享。

Abstract:

With the development of 5g and next generation communication technology, Cellular Vehicle to Everything (C-V2X) gradually presents the development trend that the application is closely combined with the new technology of mobile edge computing (MEC). However, due to the high-speed movement of vehicles, there are still a series of key technical obstacles for MEC technology to promote the rapid implementation of C-V2X network applications. In this project, MEC platform is deployed in the cellular base station which close to the vehicle, and the federated learning framework is integrated to solve the practical application of different types of services at different levels in vehicle networking. From the three aspects of model compression, condition calculation, and asynchronous algorithm design, this project attempts to introduce the deep learning framework into the "vehicle-road-cloud" service architecture, forming an adaptive collaborative mode of vertical integration and horizontal cooperation; according to the personalized requirements of vehicle users' service requests and the heterogeneity of server resources, this project proposes different scenarios of V-V, V-R, and V-R-C respectively. In order to improve the service capability of the computing environment, we propose an incentive and bidding mechanism based on game theory to optimize the competitive environment of multi-server. Finally, we enable federated learning, establish the integration scheme of block chain and federated learning. We propose a hierarchical block-chain joint learning algorithm to realize information sharing. In this way, the secure sharing of multi-attribute data in the C-V2X network is realized.

关键词 (用分号分开): 车联网; 边缘计算; 深度强化学习; 联邦学习; 区块链

Keywords (separated by;): Vehicular networks; mobile edge computing; deep reinforcement learning; federated learning; block chain



结题摘要

中文摘要（对项目的背景、主要研究内容、重要结果、关键数据及其科学意义等做简单概述）：

本项目面向蜂窝车联网（C-V2X）中车辆高速移动导致的业务需求时空快速波动、网络拓扑动态演化及多样化智能任务并发等挑战，围绕车-路-云协同服务架构、任务卸载与资源调度、激励竞争机制及隐私保护开展系统研究，致力于实现动态环境下分布式、低时延、可信的车联网智能服务。针对多层异构资源协同困难、卸载决策与资源分配耦合复杂、资源共享积极性不足以及隐私泄露与安全攻击等关键问题，形成了一系列可落地的理论与方法体系。

本项目的主要内容及重要结果包括以下几个方面：

1、自适应“车-路-云”协同服务架构与调度机制研究。面向任务实时到达、MEC能力异构及资源争用引发的效率下降，提出了基于用户偏好的分层协同卸载与迭代调度机制；进一步针对多服务提供商场景，引入边缘节点竞争与Transformer需求预测，构建“边缘竞争-云边配置”两阶段交互框架，实现动态自适应协同调度。

2、智能驱动的计算任务卸载方法研究。围绕卸载决策、卸载位置、卸载比例及资源联合分配，分别提出了负载均衡强化学习卸载（LBDDPG）、在线多DAG依赖任务调度（DPLS）与部分卸载联合优化（SIDDPG）等方法，实现时延、能耗与负载均衡综合优化，并探索全双工与无蜂窝大规模MIMO技术下的通信-计算协同优化。

3、基于博弈的激励与竞争机制研究。针对资源异构、信息不对称与隐私风险导致的共享不足与竞争失衡，提出了隐私保护的动态多属性资源竞价机制（DMRBM）等方法，构建虚拟化迁移与动态定价驱动的边云协同卸载机制，形成“竞价-定价-收益协调”的机制链条。

4、融合区块链与联邦学习的隐私保护与安全增强方法研究。提出了一种联邦学习推断攻击防护方法ASDIA，通过敏感梯度识别与校准噪声注入提升隐私-效用平衡；同时，针对导频攻击提出检测与污染信道恢复方法，并基于恢复CSI的预编码提升传输可靠性，从模型与通信层协同增强隐私与安全。

Abstract (Brief description of research background, main methods, contributions, and research data):

In cellular vehicle-to-everything (C-V2X) networks, high vehicle mobility leads to rapid fluctuations in the spatiotemporal distribution of service demands, continuously evolving network topologies, and concurrent heterogeneous intelligent tasks. To address these challenges, this project conducts systematic research on vehicle-road-cloud collaborative service architectures, task offloading and resource scheduling, incentive and competition mechanisms, and privacy preservation, aiming to enable distributed, low-latency, and trustworthy intelligent services in highly dynamic environments. Key scientific issues, including heterogeneous multi-layer resource coordination, the strong coupling between offloading decisions and resource allocation, insufficient incentives for resource sharing, and privacy leakage and security attacks during collaborative computation, are investigated, and a practical theoretical and methodological framework is established.

The main contents of this project include the following aspects:

- (1) Adaptive vehicle-road-cloud collaborative service architecture and scheduling mechanisms. Considering real-time task arrivals, heterogeneous MEC capabilities, and efficiency degradation caused by resource contention, a hierarchical collaborative offloading and iterative scheduling mechanism based on user preferences is proposed under a three-layer architecture. For multi-service-provider scenarios, edge-node competition and Transformer-based demand prediction are further introduced, and a two-stage interactive framework of “edge-side competition-cloud-edge configuration” is developed to achieve adaptive collaborative scheduling in dynamic environments.
- (2) Intelligent computation task offloading and joint resource optimization. Focusing on key issues such as whether to offload, where to offload, how to determine partial



offloading ratios, and how to jointly allocate resources, this project develops load-balancing deep reinforcement learning based offloading (LBDDPG), online scheduling for multi-DAG dependency-constrained tasks (DPLS), and partial-offloading joint optimization (SIDDPG). These methods jointly optimize latency, energy consumption, and load balancing, and further explore communication-computation co-optimization enabled by full-duplex transmission and cell-free massive MIMO.

(3) Game-theoretic incentive mechanisms and competitive resource management. To address insufficient resource-sharing motivation and unfair competition caused by resource heterogeneity, information asymmetry, and privacy risks, a privacy-preserving dynamic multi-attribute resource bidding mechanism (DMRBM) is proposed. In addition, an edge-cloud collaborative offloading mechanism driven by virtualization-based migration and dynamic pricing is designed, forming an integrated mechanism chain of “bidding-pricing-revenue coordination”.

(4) Privacy preservation and security enhancement integrating blockchain and federated learning. A federated learning inference-attack defense method (ASDIA) is proposed, which improves the privacy-utility tradeoff via sensitive-gradient identification and calibrated noise injection. To counter pilot attacks during federated learning interactions, a pilot-attack detection and contaminated-channel recovery method is developed, and transmission reliability is further enhanced using precoding based on the recovered CSI, thereby strengthening privacy and security jointly from both the learning and communication layers.

关键词（用分号分开）： 蜂窝车联网； 车-路-云协同； 任务卸载； 激励机制； 隐私保护

Keywords (separated by;): Cellular Vehicle-to-Everything (C-V2X); Vehicle-Road-Cloud Collaboration; Task Offloading; Incentive Mechanisms; Privacy Preservation



的优化模型，将卸载决策问题转化为匹配问题或多目标优化问题，并设计了基于贪婪策略与深度强化学习的卸载算法，实现了在动态环境下对任务时延和能耗的有效优化。进一步，在“车-路-云”协同场景中，针对路边侧资源不足的情况，引入云端协同卸载机制，构建分层卸载决策模型，并采用深度确定性策略梯度（DDPG）等方法，对多终端、多边缘服务器环境下的卸载决策和资源分配进行联合优化，在任务完成率、系统时延与负载均衡等方面取得显著提升。相关研究内容与申请书中提出的智能驱动任务卸载问题保持一致，并完成了模型设计与验证。

3) 基于博弈的激励与竞争机制研究。针对蜂窝车联网中多服务提供商参与协同计算、资源共享积极性不足以及资源分配不公平等问题，项目开展了基于博弈的激励与竞争机制研究。在自适应“车-路-云”协同服务架构下，构建了路边侧服务器与移动车辆之间的资源交易模型，通过引入多属性竞价机制和博弈建模方法，设计了兼顾服务提供商收益与系统整体效率的激励机制，有效缓解多边缘节点场景下的寡头效应问题，实现资源的公平竞争与高效利用。

4) 融合区块链与联邦学习的隐私保护方法研究。针对蜂窝车联网中车辆数据隐私敏感、模型参数频繁交互易受攻击的问题，项目围绕融合区块链与联邦学习的隐私保护方法开展研究。在“车-路-云”协同服务架构下，研究了面向多终端参与的联邦学习框架，通过引入区块链身份管理机制，在缺乏互信的环境中构建安全可靠的协作机制；同时，从模型更新与通信过程入手，设计了隐私保护与安全增强方法，在降低通信开销的同时提升联邦学习过程的鲁棒性与隐私保护能力。上述研究完整覆盖了申请书中提出的隐私保护关键问题。

项目执行人在整个项目执行期间，以该项目研究内容为导向，发表科技论文 22 篇，其中 CCF A 类/SCI 一区期刊论文 9 篇，CCF B 类/SCI 二区期刊 6 篇，授权国家发明专利 5 项，新申请国家发明专利 10 项，科研工作相关奖励 3 项，指导在读硕士研究生 13 人，在读博士研究生 3 人，项目组已培养毕业硕士 13 人，资助学生参与国内学术会议 20 余次。

(2) 研究目标完成情况。